SESSION 07 - MODEL CARD AND DATASET DATASHEET

By: Margorie Fernanda Pérez Simba

Scope. This report documents the proposed student dropout prediction framework and the preliminary institutional data available for the doctoral research. The definitive thesis dataset has not yet been constructed, and no confidential student records are included in this report or shared publicly.

1. MODEL CARD

Model details

- **Name / version:** Explainable, Multimodal, and Temporal Student Dropout Prediction Framework v0.1 (proposed)
- **Owner / contact:** Margorie Fernanda Pérez Simba
- **Date:** 2026-10
- **Development status:** Research design stage. The definitive model has not yet been trained or validated, and no operational version has been deployed.
- **Architecture:** Proposed explainable, multimodal, and temporal AI framework combining structured institutional variables, temporal patterns, and authorized textual information. Candidate algorithms and component designs will be compared during the doctoral study rather than assumed in advance.
- **License:** Research documentation only. The proposed model has not been publicly released.

Intended use

- **Primary intended uses:** If adequately validated, support the early identification of students who may be at risk of dropout and help authorized institutional teams prioritize timely, preventive retention support.
- **Out-of-scope / misuse:** The model must not be used to make automatic decisions about admission, financial aid, academic sanctions, or student exclusion. It must not replace professional judgment or be applied to other institutions without external validation.
- **Intended users:** Authorized staff responsible for student retention, academic advising, institutional analytics, and academic decision support.

Factors

- **Relevant groups / conditions:** Model performance may vary across academic colleges, degree programs, admission cohorts, study stages, and student groups with different academic or financial conditions. Changes of degree program within the same institution are excluded from the dropout outcome.
- **Evaluation factors:** Performance should be reported for the complete institutional population and disaggregated by academic college, degree program, admission cohort, and other relevant subgroups when the available sample size permits a reliable evaluation.

Metrics

- **Performance measures:** ROC-AUC, precision-recall AUC, recall, precision, F1-score, balanced accuracy, confusion matrix, and probability calibration. Recall for the dropout class is especially important because false negatives represent students at risk who may not receive timely support.
- **Decision thresholds:** No operational threshold has been selected. Candidate thresholds will be evaluated using validation data and aligned with the institution's intervention capacity and the relative consequences of false negatives and false positives.
- **Variation approach:** Performance will be reported across multiple predefined random seeds or repeated validation runs using the mean, standard deviation, and 95% confidence intervals when applicable. Temporal variation across cohorts will also be examined.

Evaluation data

- **Datasets:** The definitive evaluation dataset has not yet been constructed. The planned design will reserve data that are not used for model training or hyperparameter tuning and will include later academic cohorts for temporal validation when the available outcome window permits.
- **Preprocessing:** The planned pipeline will remove or pseudonymize direct identifiers. Missing-value treatment, categorical encoding, feature selection, scaling when required, and any class-balancing procedure will be fitted using training data only and then applied to validation and test data. Only information available before the defined prediction point will be used to prevent data leakage.

Training data

- **Datasets:** A confidential preliminary extract of real historical student records from one higher education institution in Ecuador is available to assess data availability and inform the research design. It is not the definitive thesis dataset and has not been used to claim final model performance. The definitive dataset will be constructed after the scope, prediction point, eligible population, authorized variables, and available cohorts are finalized.

- **Outcome definition:** Dropout is determined from institutional registration records according to the four regular academic periods following the student's first registration. A student who changes degree program but remains registered at the same institution is not classified as a dropout.
- **Known limitations:** The preliminary extract does not yet represent the final study population, predictor set, or cohort range. The definitive data will come from a single institution and may reflect its specific academic policies, student population, data quality, and historical practices. Some relevant circumstances may not be recorded, and class imbalance or changes across periods may affect performance.

Quantitative analyses

- **Overall results:** UNKNOWN: no definitive thesis model has been trained or validated. Final results will be reported only after the dataset and validation protocol are finalized and will include ROC-AUC, precision-recall AUC, recall, precision, F1-score, balanced accuracy, calibration, and the confusion matrix.
- **Disaggregated results:** UNKNOWN: subgroup evaluation has not yet been conducted. Performance will be disaggregated by academic college, degree program, admission cohort, and other relevant groups when sample sizes permit reliable reporting.
- **Model interpretation:** UNKNOWN: no definitive model explanations are available. The planned evaluation will compare global and local explanation methods, including SHAP when appropriate, and assess whether explanations are useful to authorized institutional users.

Ethical considerations

- **Risks:** The proposed model would process confidential student information and could reproduce historical inequalities present in institutional data. Incorrect predictions could stigmatize students, create unnecessary interventions, or fail to identify students who need support. Users might also place excessive confidence in predicted probabilities.
- **Mitigations:** Use de-identified or pseudonymized data, restrict access to authorized personnel, apply institutional data-protection procedures, evaluate performance and fairness across relevant subgroups, provide explanations for predictions, maintain human oversight, and prohibit punitive or fully automated decisions based on model outputs.

Caveats and recommendations

- **Caveats:** This document describes a proposed system, not a completed or deployable model. Future predictions will be based on historical data from one institution and will not establish causal relationships. A high predicted probability will not mean that a student will necessarily drop out, and performance may change over time because of data drift.
- **Recommendations:** Do not deploy the system until the definitive dataset, prediction point, validation protocol, fairness assessment, calibration, explainability analysis, and governance requirements have been completed. If validated, use it only as decision support for preventive student services with human review, temporal monitoring, and documented model updates.

2. DATASET DATASHEET

Motivation

- **Purpose:** The source records were originally created for routine academic and administrative processes. A confidential preliminary extract of these real institutional records is currently available to assess data availability, refine the research scope, and design the future thesis dataset. The definitive analytical dataset has not yet been constructed.
- **Created by / funded by:** The source records were generated by the academic and administrative units of one higher education institution. Margorie Fernanda Pérez Simba will construct the definitive research dataset from authorized institutional sources for doctoral research. Funding information: UNKNOWN, to be confirmed.

Composition

- **What instances represent:** The preliminary extract contains historical student records across academic periods and may include multiple operational records for the same student. The definitive unit of analysis will be established after the prediction point and longitudinal representation are finalized; it may be one student at a defined prediction point or an authorized temporal sequence of student records.
- **Variables:** The preliminary extract contains real structured academic, administrative, contextual, and temporal fields. The definitive dataset may incorporate additional authorized variables, newer cohorts, and textual sources required by the final research scope. The final record count and predictor list are therefore not yet fixed.
- **Labels:** The planned binary outcome is dropout or continued enrolment. Dropout will be derived from institutional registration records across the four regular academic periods following the student's first registration. A change of degree program within the same institution will not be labeled as dropout.
- **Sample vs population:** The preliminary extract contains selected historical cohorts from one higher education institution in Ecuador and is not the definitive thesis sample. Final inclusion and exclusion criteria will be applied when the analytical dataset is constructed. Neither the current nor future institutional dataset represents all higher education students in Ecuador or other institutions.

- **Missing data:** Operational records may contain missing or incomplete values. Their amount, pattern, possible meaning, and treatment have not yet been finalized and will be documented for each version of the definitive analytical dataset.

Collection process

- **How collected:** The source data were recorded through routine institutional processes such as admission, registration, academic progression, and student support. A preliminary extract is available through the researcher's institutional work context. Construction and use of the definitive thesis dataset will require the corresponding institutional authorization and data-governance procedures.
- **Timeframe:** The preliminary extract contains historical cohorts and multiple regular academic periods. The final timeframe is not fixed because it will depend on when model development begins, which cohorts meet the eligibility criteria, and whether the four-period outcome window is complete. Newer cohorts may be added when eligible.
- **Consent / ethics:** The records were collected for institutional academic and administrative purposes rather than specifically for this study. Their secondary use for doctoral research requires formal institutional authorization, confidentiality safeguards, and ethics review when applicable. Final approval status: UNKNOWN, to be confirmed before definitive data preparation or model training.

Preprocessing / cleaning / labeling

- **Steps:** No final preprocessing pipeline has been applied as part of this course documentation. The planned procedure will remove or pseudonymize direct identifiers, check duplicates and invalid values, document missingness, construct the dropout label from registration records, and define the prediction point before modeling. Data will be split by the planned validation design before feature selection, imputation, encoding, scaling, or class balancing to prevent data leakage.
- **Raw data available:** The raw institutional records are not publicly available because they contain confidential student information. Access is restricted to authorized personnel under institutional data-protection procedures.

Uses

- **Appropriate uses:** At this stage, the preliminary extract may be used under institutional authorization to understand data structure, availability, and quality and to refine the research design. Once the definitive dataset and approvals are in place, appropriate uses will include student dropout prediction research, evaluation of explainability and fairness, analysis of temporal patterns, and development of preventive decision support under human supervision.
- **Should NOT be used for:** The dataset should not be used for public identification of students, punitive decisions, automatic denial of admission or financial aid, academic sanctions, commercial profiling, or research unrelated to the approved purpose. It should not be used to make claims about other institutions or national student populations without additional validation.

Distribution

- **How distributed:** Individual-level student records are confidential and will not be uploaded to GitHub or another public repository. Access will remain limited to authorized personnel in controlled environments. Only non-identifying metadata, aggregated results, or institutionally approved examples may be shared publicly.
- **License / restrictions:** The dataset does not have an open-data license. Its use is subject to institutional authorization, confidentiality requirements, data-protection policies, and any applicable ethics approval. The specific data-use agreement and retention conditions are UNKNOWN and must be formally confirmed.

Maintenance

- **Maintainer:** The institutional units responsible for the source information systems remain the data stewards. Any authorized research copy and its documentation will be managed by Margorie Fernanda Pérez Simba in coordination with the relevant institutional units.
- **Update plan:** The definitive dataset may incorporate additional authorized variables, newer academic periods, and eligible cohorts as the doctoral scope is finalized. Each research version will document its extraction date, included cohorts, eligibility rules, variable definitions, prediction point, outcome construction, and preprocessing changes. Data quality, subgroup representation, and temporal drift will be reviewed after each update.